



Health Services Department

Public Health Division

Donald R. Rowe, Director

Thomas L. Charron, M.D., M.
Health Officer
Assistant Director

EMERGENCY MEDICAL SERVICES

POLICY MEMORANDUM #5600

EFFECTIVE DATE: 11/01/91

APPROVED BY:

Handwritten signature of Thomas L. Charron.

THOMAS L. CHARRON, M.D., F.P.S., SOLANO COUNTY HEALTH OFFICER, EMS MEDICAL DIRECTOR

REVIEWED BY:

Handwritten signature of Steven J. Kanitschke.

STEVEN J. KANITSCHKE, M.D., F.A.C.E.P., ASSISTANT EMS MEDICAL DIRECTOR

SUBJECT: EQUIPMENT & SUPPLIES FOR BLS AMBULANCE UNITS

AUTHORITY: CALIFORNIA CODE OF REGULATIONS, TITLE 13, CHAPTER 5, Section 1103.2.

PURPOSE/POLICY:

BLS Ambulances operating in Solano County shall be equipped as outlined in Attachment A of this Policy in addition to the California Code of Regulations, Title 13 Requirements.

Although it is the intent of Solano County EMS to have ALS Ambulances respond to all Emergency and 911 requests, it is recognized that sick and injured patients are transported by BLS Units. These Units shall be equipped and supplied so that EMT-1s functioning on these Units can appropriately care for patients to whom they respond or accidentally encounter.

All BLS Ambulances shall maintain the standard stocking equipment and supplies in Attachment A. BLS units may remain in service with the minimal stock listing for no more than 48 hours. BLS units which do not have the minimal required equipment and/or supplies shall be removed from active service until re-supplied.

ITEM	Quantity STANDARD Stocking	Quantity MINIMAL Stocking
☐ An ambulance cot & collapsible stretcher; or two stretchers, one of which is collapsible	1	1
☐ Straps to secure the patient to the stretcher or ambulance cot, and means of securing the stretcher or ambulance cot in the vehicle	2	2
☐ Clean blankets	2	2
☐ Pillows	2	2
☐ Clean Pillowcases	3	3
☐ Clean Sheets	6	6
☐ Stethoscopes	2	1
☐ Adult Sphygmomanometers	2	1
☐ Pediatric Sphygmomanometers	1	1
☐ Small flashlight, or penlight	1	1
☐ Alcohol wipes	20	20
☐ Bite sticks, either commercial or made of three tongue blades taped together & padded.	2	1
☐ Sterile obstetrical supplies including as a minimum: gloves, umbilical cord tape or clamps, dressings, towels, bulb syringe, and clean plastic bags. ALTERNATIVE: Commercially prepared OB Kit.	1	1
☐ Bandage Shears	1	1
☐ Band-aids in assorted sizes.	20	5
☐ Sterile gauze pads, 4" x 4", packaged individually	24	10
☐ Combine dressings (Surgi-pad type), packaged individually, approximately 5" x 9"	8	5
☐ Universal dressings (ABD's multi-trauma dressings), approximately 10" x 36", compactly folded and packaged in a convenient size	6	3
☐ Soft-roller, self-adhering bandages, 4" x 5 yds, or 3" x 5 yds.	4	2
☐ Soft-roller, self-adhering bandages, 6" x 5 yds.	6	2
☐ Petroleum gauze or other commercially prepared occlusive dressings (ALTERNATIVE: roll of aluminum foil, 18" x 25", sterilized and wrapped).	2	2
☐ Rolls adhesive tape or other acceptable substitute, ½ " wide	3	1
☐ Rolls adhesive tape or other acceptable substitute, 1" wide	3	1
☐ Rolls adhesive tape or other acceptable substitute, 3" wide	2	1
☐ Cold packs	4	1
☐ Triangular bandages, 42" wide	4	2
☐ Large safety pins	8	2
☐ Sterile burn sheets	2	2
☐ Ankle and wrist restraint – soft ties are acceptable (leather restraints are recommended)*	1 set	1 set
☐ Gallon or more of potable water in a covered, secured plastic container	1	1
☐ Disposable drinking cups	6	2

ITEM	Quantity STANDARD Stocking	Quantity MINIMAL Stocking
☐ Bedpan, fracture pan	1	1
☐ Urinal	1	1
☐ Rigid spine boards, approximately 14" or more wide by 72" long, each with two sets of straps for immobilization of suspected cervical or spinal injuries	2	1
☐ Rigid spine board, approximately 14" wide by 32" long, or acceptable substitute, each with 2 sets of straps	1	1
☐ Cervical collars: 1 small, 2 medium & 1 large (Stiff – sizes to fit all patients over one year old approved by the EMS Medical Director)	4	4
☐ Sandbags, loosely filled, or equivalent material to restrict movement	4	2
☐ Adult size half-ring traction splint for lower extremity with limb support slings, padded ankle hitch traction strap, and heel rest or an equivalent device	1	1
☐ Pediatric size half-ring traction splint for lower extremity with limb support slings, padded ankle hitch traction strap, and heel rest or an equivalent device	1	0
☐ At least three 3" x 15" arm splints and three 3" x 36" or longer leg and thigh splints, with a soft or cushioned surface or roll material for padding, for immobilizing and protecting fractured extremities. Equivalent padded board, wrap-around, wire ladder, inflatable, or cardboard splints will meet this requirement	3	3
☐ 3 E or D-cylinders of oxygen (USP) with one O ₂ regulator/flow-meter (portability for two O ₂ cylinders required)	3	3
☐ Hand-operated, self-inflating bag-valve-mask unit with a bag capacity of 1600 ml, with a reservoir that allows the administration of 100% O ₂ with clear face masks in adult and child sizes.	1	1
☐ Pediatric size hand-operated, self-inflating bag-valve-mask unit which is capable of use with oxygen , with clear face masks in child and infant sizes without pop-off valve.	1	1
☐ Nasal cannulas, adult size	4	2
☐ Simple face masks, pediatric size	2	1
☐ Oxygen supply tubing	3	1
☐ Non-rebreathing face masks, adult size	2	1
☐ Oropharyngeal airways in assorted sizes for infants, children and adults (Sizes 1 – 6)	1 set	1 set
☐ Nasopharyngeal airways in assorted sizes	3	3
☐ Portable suction apparatus capable of a vacuum of up to 25mm of O ₂ mercury, with wide-boring tubing, and rigid pharyngeal suction tip, and small, medium and large suction catheters.	1	1
☐ Tube K-Y gel or other commercially prepared lubricating gel	1	1
☐ Pediatric port-a-warm, neonatal warmer, or approved equivalent	1	1
☐ Fox shields or protective metal eye patches	1	1
☐ Maps of the entire count (coded with grid system as prescribed by the EMS Agency)	1 set	1 set
☐ Triage Tags and Disaster Materials as required by the County MCI Plan	20 tags	5 tags
☐ 1 box of non-sterile disposable gloves	1	1

SUBJECT: ATTACHMENT A – REQUIRED EQUIPMENT/SUPPLIES
For BASIC LIFE SUPPORT AMBULANCE UNITS

POLICY: 5600
DATE: 11/01/91

ITEM	Quantity STANDARD Stocking	Quantity MINIMAL Stocking
☐ <u>HAZ-MAT Kit</u> 10 large trash bags 16 Chucks 4 ea disposable gowns, slippers 4 ea surgical masks 1 large container of Vaseline 1 medium bottle dish soap 2 pair splash goggles 6 pair heavy duty gloves	1 kit	1 kit
☐ Communication equipment required by the EMS Agency to link with central dispatch (to take effect on or before implementation date of a central dispatch center).	1 set	1 set
☐ Additional supplies and equipment as required by the California Highway Patrol as required by Title 13, California Administrative Code	As required	As required

Solano County Health & Social Services Department



Mental Health Services
Public Health Services
Substance Abuse Services
Older & Disabled Adult Services

Eligibility Services
Employment Services
Children's Services
Administrative Services

Patrick O. Duterte, Director

EMERGENCY MEDICAL SERVICES AGENCY

Michael A. Frenn
EMS Agency Administrator

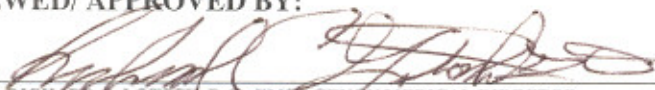
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Richard C. Lotsch, D. O.
EMS Agency Medical Director

POLICY MEMORANDUM 5700

20 May 2008

REVIEWED/ APPROVED BY:


RICHARD C. LOTSCH, D.O., EMS AGENCY MEDICAL DIRECTOR


MICHAEL FRENN, EMS AGENCY ADMINISTRATOR

SUBJECT: EQUIPMENT, SUPPLIES & MEDICATIONS – ALS UNITS

AUTHORITY: CALIFORNIA CODE OF REGULATIONS, TITLE 13, CHAPTER 5, SECTION 1103.2

PURPOSE/POLICY:

All Solano County ALS units shall maintain minimum standard stocking equipment and supplies. The attached list shall be considered minimal stocking. ALS units that do not have the minimum required equipment and supplies are to be removed from active service until re-supplied. A provider may choose to stock a unit with additional quantities as necessary.

- I. **ALL ALS TRANSPORTING UNITS** shall be equipped with safety and emergency equipment required for ambulances by the California Vehicle Code and the California Code of Regulations and regulations promulgated thereunder.
- II. **ALL ALS NONTRANSPORTING UNITS** shall be equipped with the minimum stock shown in Attachment A, with the exception of a gurney and linen.
- III. **ALL ALS UNIT CURRENTLY APPROVED AS ALS PROVIDERS** shall be required to demonstrate compliance with equipment and supplies. Currently approved ALS Units of agencies which are exempt from permit requirements shall be required to demonstrate compliance with these minimum levels of stocking.

IV. INITIAL MEDICATION STOCK:

- A. The Solano County EMS Medical Director or the Provider Medical Director shall issue a prescription for the medications, solutions and supplies listed on Attachment A as needed for the initial stocking of each unit authorized as an ALS unit. No pharmaceutical or related supplies other than those listed shall be carried.
- B. No drugs, medications, needles or catheters are to be located on a vehicle which is not totally equipped, staffed and approved to provide ALS services in Solano County.

V. RE-STOCK:

- A. Each ALS provider shall have a written policy outlining their re-stocking procedure. It is the responsibility of the ALS provider to re-stock their units unless other arrangements have been made.
- B. No agency shall bill a patient for medications or associated supplies if the supplies were replaced by a hospital.

VI. NARCOTICS

- A. Each ALS Provider shall be responsible for maintaining narcotic tracking in compliance with the Federal Drug Enforcement Administration Regulations.
- B. Each ALS provider shall be responsible for quality assurance as outlined in Policy #2203.

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POLICY MEMORANDUM 5700

NON-TRANSPORT UNIT STOCK

Medication/Supply & Concentration Size (where applicable)	ALS NON-TRANSPORT Minimum Stock	BLS NON-TRANSPORT Minimum Stock
☐ Acetaminophen	1 bottle	0
☐ Albuterol; 0.84% solution	3	0
☐ Adenosine; 6mg	5	0
☐ Amiodarone; 150mg	2	0
☐ Aspirin; 81mg chewable	1 bottle	1 bottle
☐ Atropine Inj; 1mg	2	0
☐ Atropine Inj; 8mg (.4mg/ml)	2	0
☐ Calcium Chloride Inj. 10%; 1gm	1	0
☐ Dextrose 50% Inj; 25gm	1	0
☐ Dextrose 10% Inj	1	0
☐ Diphenhydramine (Benadryl) Inj.; 50mg	2	0
☐ Epinephrine Inj; 1:1,000-1ml	1	0
☐ Epinephrine Inj; 1:10,000 – 10ml	6	0
☐ Epinephrine Auto-Injector	0	1 Adult 1 Pediatric
☐ Fentanyl or Morphine;	2	0
☐ Ipratropium Bromide; 0.5mg/2.5mL (OPTIONAL)	3	0
☐ Glucagon; 1 unit (1.0mg)	1	0
☐ Glucose; Paste	1	1
☐ Lidocaine Inj; 100mg	4	0
☐ Midazolam (Versed) 5mg/2ml	2	0
☐ Morphine Sulfate; 10mg (or Fentanyl)	2	0
☐ Nitroglycerine; 0.4mg/tablet or spray	1 bottle	0
☐ Naloxone (Narcan) Inj.; 2mg	2	2 naloxone prefilled kits
☐ Ondansetron; 4mg/2mL or equivalent ODT	2	0
☐ Sodium Bicarbonate 8.4%; 50 mEq/50ml	1	0
☐ Pediatric Sodium Bicarbonate 4.2%; 5 mEq/10ml	1	0

Medication/Supply	ALS NON- TRANSPORT Minimum Stock	BLS NON- TRANSPORT Minimum Stock
☐ Normal Saline IV bags; 1,000mL	2	0
☐ IV Administration sets	2	0
☐ T-Connector (for Adenosine Adm. at IV Hub)	2	0
☐ 3-way stopcock/extension tubing	2	0
☐ 22g IV catheters (catheter-over-needle type)	2	0
☐ 20g IV catheters (catheter-over-needle type)	2	0
☐ 18g IV catheters (catheter-over-needle type)	2	0
☐ 16g IV catheters (catheter-over-needle type)	2	0
☐ 14g IV catheters (catheter-over-needle type)	2	0
☐ 10 or 12cc syringes with needle	2	0
☐ 5 or 6cc syringes with needle	2	0
☐ 3cc syringes with needle	2	0
☐ 1cc syringes with needle	2	0
☐ 18 gauge needles, 1-1/2"	2	0
☐ Sodium Chloride for irrigation (1000 ml, sterile)	1	1
☐ Hand-held disposable nebulizers	2	0
☐ 30 or 50 cc syringes w/luer-lock hub to perform IO Fluid Challenge	2	0
☐ Pediatric (Broselow first five minutes or reasonable equivalent) tapes for calculating Pediatric Medication Dosages	1	0
☐ Saline Lock supplies	2	0
☐ Mucosal Atomizer Device	2	2 If not already included with naloxone kits

Medication/Supply	ALS NON-TRANSPORT Minimum Stock	BLS NON-TRANSPORT Minimum Stock
☐ Portable, battery powered Monitor-Defibrillator with print-out, pacing, 12 Lead EKG, waveform capnography, and cardioversion capabilities ☐ Adult and pediatric defibrillation paddles or “hands-free” equivalent ☐ Spare batteries ☐ Sets of leads ☐ Spare roll of EKG paper ☐ Sets of electrodes ☐ Sets of defibrillator pads or 1 tube of defibrillator gel	 1 1 1 2 1 3 1	 1 AED with appropriate cables and defibrillation pads
☐ 10 channel UHF paramedic radio which is approved by the Solano County Department of Communications	1	1
☐ Stethoscopes	1	1
☐ Adult/pediatric sphygmomanometers	1	1
☐ Small flashlight or penlight	1	1
☐ IV labels for adding medications	4	0
☐ Pair sterile gloves	2	2
☐ IV tourniquets	2	0
☐ CAT tourniquets	3	3
☐ Alcohol wipes	20	20
☐ Sterile obstetrical supplies including, as a minimum: gloves, umbilical cord tape or clamps, dressings, towels, bulb syringe, and lean plastic bags	1	1
☐ Bandage shears	1	1
☐ Band-aids in assorted sizes	5	5
☐ Sterile gauze pads, 4” x 4”, packaged individually	6	6
☐ Combine dressings (Surgipad type), packaged individually, approximately 5” x 9”	5	5
☐ Universal dressings (ABD’s multi-trauma dressings), approximately 10” x 36”	3	3
☐ Soft-roller, self-adhering bandages, 3” or 4” x 5 yds.	2	2
☐ Soft-roller, self-adhering bandages, 6” x 5 yds.	2	2
☐ Petroleum gauze or other commercially prepared occlusive dressings	1	1
☐ Rolls adhesive tape or other acceptable substitute, ½”, 1”, 3”	1	1
☐ Cold packs	1	1
☐ Triangular bandages, 40” wide	2	2
☐ Sterile burn sheets	1	1
☐ Hemostatic dressing	3	3

Medication/Supply	ALS NON-TRANSPORT Minimum Stock	BLS NON-TRANSPORT Minimum Stock
☐ Rigid spine boards, at least 14" wide by 72" long, each with two sets of straps	1	1
☐ Rigid spine board, approximately 14" wide by 32" long, or acceptable substitute, each with 2 sets of straps	1	1
☐ Cervical collars	1 each size	1 each size
☐ Neck blocks or equivalent device to restrict movement	1	1
☐ Adult & pediatric lower extremity traction splint	1	1
☐ 3" x 15" arm splints and three 3" x 36" or longer leg and thigh splints, with a soft or cushioned surface or roll material for padding, for immobilizing and protecting fractured extremities. Equivalent padded board, wrap-around, wire ladder, inflatable, or cardboard splints will meet this requirement	1 each	1 each
☐ E or D cylinders of oxygen (USP) with one regulator/flow-meter, at least 1 must be portable	1	1
☐ Hand-operated, self-inflating bag-valve-mask unit with a bag capacity of at least 1600ml, with a reservoir that allows the administration of 100% oxygen with clear face masks in adult and child sizes	1	1
☐ Pediatric size hand-operated, self-inflating bag-valve-mask unit which is capable of use with oxygen, with clear face masks in child and infant sizes, without pop-off valve	1	1
☐ Infant warming device	1	1
☐ Nasal cannulas, adult and pediatric sizes	2	2
☐ Face masks, non-rebreathing, adult and pediatric sizes	2	2
☐ Oropharyngeal airways in assorted sizes for infants, children & adults (sizes 1 – 6)	1	1
☐ Nasopharyngeal airways	1 set	1 set
☐ Portable suction apparatus capable of a vacuum of up to 25 mm of mercury, with wide-bore tubing, and a rigid pharyngeal suction tip, and small, medium, and large suction catheters	1	1
☐ Bulb syringe	1	1
☐ Tube of commercially prepared lubricating gel	1	1
☐ Infrared/Tympanic Thermometer or equivalent	1	1

Medication/Supply	ALS NON-TRANSPORT Minimum Stock	BLS NON-TRANSPORT Minimum Stock
☐ Cuffed endotracheal tubes (11 total to a set) ☐ 2 each of the following sizes: 7.0, 7.5, 8.0 ☐ 1 each of the following sizes: 6.0, 6.5	2 each 1	0
☐ Endotracheal tube tie downs or holders	1	0
☐ Malleable, round-tipped stylet	1	0
☐ Laryngoscope handle with light with: ☐ Adult straight #4 blade ☐ Adult curved #3 blade ☐ Infant straight #1 blade ☐ Infant curved #2 blade ☐ spare bulb ☐ spare batteries	1 1 1 1 1 1 1	0
☐ Magill forceps – adult and pediatric sizes	1	0
☐ End tidal CO ₂ monitor	1	0
☐ Esophageal detection device (bulb or syringe type)	1	0
☐ Eye protection shields	2	2
☐ Maps of the entire County (with Thomas Brother Grids)	1	1
☐ Triage tags and disaster materials as required by the County MCI plan	20	20
☐ Non-sterile disposable gloves	1 box each size	1 box each size
☐ Thoracostomy Kit (or approved equivalent for chest decompression)	1	0
☐ Needle cricothyrotomy kit (or approved equivalent)	1	0
☐ Container to dispose of needles and other medical waste	1	1
☐ Other equipment authorized by the EMS Medical Director ☐ King Airway (1 set of 3) ☐ CPAP devices ☐ EZ IO ☐ Glucose monitoring device ☐ Pulse oximetry monitor ☐ External cardiac pacing equipment	<u>1</u> <u>1</u> <u>1</u> 1 1 1	1 (If provider is authorized) 1 0 1 1 0

Solano County Health & Social Services Department

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EMERGENCY MEDICAL SERVICES AGENCY

Richard C. Lotsch, D.O.
EMS Agency Medical Director

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Ted Selby
EMS Agency Administrator

POLICY MEMORANDUM 5800

DATE: September 15, 2010

REVIEWED/APPROVED BY:


RICHARD C. LOTSCH, D.O., EMS AGENCY MEDICAL DIRECTOR


TED SELBY, EMS AGENCY ADMINISTRATOR

SUBJECT: USE OF EMS AIRCRAFT

AUTHORITY: CALIFORNIA ADMINISTRATIVE CODE, TITLE 22, CHAPTER 6, SECTIONS
100206 – 100228.

PURPOSE/POLICY:

This policy outlines indications for the use of Emergency Medical Services (EMS) aircraft, methods and procedures for dispatch, and assigns responsibility for EMS aircraft destination decisions and for communication with the receiving facility. EMS aircraft are intended to be used in circumstances when patients with traumatic injuries require rapid transport to a trauma center, in accordance with the Solano County Trauma Triage Protocol.

I. DISPATCH

- A. Dispatch of EMS Aircraft (911 calls) within Solano County shall be initiated by the County's EMS Aircraft "Designated Dispatch Center" (currently Solano County Sheriff's Dispatch Center). The closest available aircraft will be dispatched to the scene. During call scenarios where there is more than one aircraft provider available, a daily dispatch rotation will be applied. This aircraft rotation will occur when aircraft are dispatched from Buchanan Field in Concord. Requests for dispatch of an EMS Aircraft will originate with the incident commander following an assessment which takes into consideration the patient's medical condition along with logistical factors:
 1. Scene safety, which requires a landing zone (LZ) which is located within reasonable proximity and is safe for both air and ground crews.

2. Patient weight within aircraft limits, non-combative, free of hazardous materials or fumes.
 3. EMS aircraft availability within a time frame which would be estimated to result in a marginally shorter arrival time at the trauma center Emergency Department (ED) of at least 10 minutes over ground response and transport.
 4. Medical needs which are either particularly time sensitive or requiring the specialized skills of a flight nurse:
 - a. Deteriorating mental status;
 - b. Unstable vital signs;
 - c. Potential for unstable airway;
 - d. Amputations, excluding digits, with potential for reattachment.
- B. The Dispatcher's decision regarding which aircraft to send shall be in accordance with the rotational protocol established by the EMS Agency (Attachment A).
- C. Solano County Dispatch will provide the following to the aircraft dispatch:
1. Location;
 2. Requesting Agency;
 3. Nature of Request.
- D. Aircraft dispatch should contact the Requesting Dispatch Agency to obtain additional information which may include:
1. Number of patients, type and extent of injuries or illness, estimated weight;
 2. Landing site information (if possible);
 3. Coordinates;
 4. Landmarks identifiable from the air;
 5. Terrain and obstacles (e.g., wires & poles in the immediate area);
 6. On-scene Unit Designator;
 7. Significant on-scene weather conditions.

II. CRITERIA FOR SIMULTANEOUS DISPATCH

- A. No ALS Ground Ambulance immediately available such as in some of the remote locations in Solano County.
- B. The Dispatcher at the designated EMS Aircraft Dispatch Center should consider the following criteria in making the decision to simultaneously dispatch EMS aircraft.

INJURY CRITERIA	Motor Vehicle Accident (MVA) CRITERIA
➤ Advanced Life Support (ALS) Ground ambulance response >20 min; ➤ Victim is unconscious, due to trauma; ➤ Penetrating injury chest/abdomen/head/neck/groin/buttock; ➤ Falls of ≥ 15 feet; ➤ Explosion or blast injuries; ➤ Partial/full amputation, excluding digits; ➤ Major Electrocution (lightning strike, high voltage)	➤ ALS Ground ambulance response >20 min. ➤ Unrestrained, crash speed >35mph; ➤ Vehicle rollover; ➤ Victim entrapment; ➤ Victim ejection; ➤ Motorcycle victim ejected speed >20mph; ➤ Pedestrian struck by vehicle speed >20mph ➤ Multiple victims.

III. LANDING SITES

- A. There are three (3) types of landing sites to be utilized by EMS aircraft:
 - 1. Sites chosen at or near the scene of an incident;
 - 2. Pre-designated landing sites.
 - 3. Heliports and/or airports.

IV. COMMUNICATIONS

- A. Equipment:
All EMS aircraft will be equipped, at a minimum, with radio equipment approved by the EMS Medical Director upon recommendation of the Director of Solano County Communications Department. At a minimum, EMS Aircraft must have the capability of communicating with Solano County Central Dispatch and ground units using Fire I, CALCORD 1 and CALCORD 2.
- B. Channels:
EMS Aircraft Providers shall utilize and maintain medical communications with local EMS first responders, EMS providers, and Base Hospitals as specified by the EMS Agency.
- C. Protocol:
 - 1. Requests for EMS Aircraft will be made from responding units to their respective Dispatch Center.
 - 2. Dispatch centers will notify Solano Dispatch of the need for service and include:
 - a. Location of call;
 - b. Nature of call;
 - c. On-scene unit designator.
 - 3. Once enroute, EMS Aircraft Dispatch will notify the requesting agency dispatch (not Solano Sheriff's Office).
 - 4. As soon as possible, responding aircraft will utilize the following communications:
 - a. FIRE 1 for radio communications while enroute;
 - b. CALCORD for on-scene communication;
 - c. FIRE 2 if CALCORD is unavailable.

V. CANCELLATION OF RESPONSE

EMS Aircraft Response may be cancelled by the most medically qualified on-scene EMS personnel in conjunction with the scene commander when it is determined that:

- A. The patient does not meet trauma triage criteria and/or their clinical condition doesn't warrant EMS Aircraft transport;
- B. Ground ambulance transportation is more appropriate (e.g. because of delay in air ambulance response, proximity of a receiving hospital, etc.);
- C. A patient refuses to be transported by air ambulance;
- D. A patient refuses medical aid and/or transportation;
- E. Transport of the patient presents a danger to the flight crew;
- F. Pilot determination based on flight operations and/or environmental conditions.

VI. EMS AIRCRAFT OPERATIONAL DISPATCH

- A. Each authorized EMS air ambulance/rescue service shall maintain an operational dispatch center 24 hours/day; 7days/week. The dispatch center shall be able to immediately confirm the provider's ability to respond to an EMS incident upon notification by Solano County Dispatch Center. The EMS aircraft operational dispatch center will monitor the status of area receiving trauma facilities. The dispatch center and the responding aircraft will jointly determine the closest (as measured by flying time) available trauma facility, and will communicate patient information to the destination facility.
- B. Operational Log
The EMS aircraft service shall maintain a log and/or records of all EMS aircraft requests from all requests for EMS aircraft originating in Solano County. The minimum data to be recorded and submitted to the Agency on a monthly basis and shall include:
1. Date;
 2. Time of notification by the Solano County Dispatch Center;
 3. Time the EMS aircraft is enroute;
 4. Time of arrival at scene;
 5. Time of departure from scene;
 6. Time of arrival at receiving facility;
 7. Time available and/or time call was terminated when services were not needed;
 8. Entity requesting the service and/or cancelling the service;
 9. Pertinent factors associated with terminated requests.
- C. EMS Aircraft shall comply with all requirements specified in applicable agreements, contracts, policies, and procedures issued by the EMS Agency.

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HELICOPTER ROTATION SCHEDULE

ZONE 1	Fire districts include:
Ryer Island,	Isleton FD Isleton Fire Protection District
AIR AMBULANCE CALL ORDER	
CALSTAR 8 – Vacaville, Closest Air Ship	
CALSTAR 1 – Concord, Rotation	
REACH 3 – Concord, Rotation	
REACH 2 – Stockton	
CALSTAR 3 – Auburn	
CHP – H30	

ZONE 2	Fire Districts include:
Dixon Fire District	City of Vacaville Vacaville Fire District
City of Dixon	Fairfield Montezuma
Rio Vista	Suisun District City of Suisun
Cordelia	CDF west of Cordelia
AIR AMBULANCE CALL ORDER	
CALSTAR 8 – Vacaville, Closest Air Ship	
CALSTAR 1 – Concord, Rotation	
REACH 3 – Concord, Rotation	
REACH 2 – Stockton	
Reach 1 – Santa Rosa	
CHP – H30	

ZONE 3	Fire districts include:
Vallejo	Benicia Delta
AIR AMBULANCE CALL ORDER	
CALSTAR 1 – Concord, Rotation	
REACH 3 – Concord, Rotation	
CALSTAR 8 – Vacaville	
REACH 1 – Santa Rosa	
REACH 2 – Stockton	
CHP – H30	



Health & Social Services Department

Donald R. Rowe, Director

Public Health Division - Mental Health - Public Guardian
Adult and Child Services - Eligibility & Employment Services
Substance Abuse Services

Public Health Division
Thomas L. Charron, M.D., M.P.H.
Health Officer/EMS Medical Director
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POLICY MEMORANDUM #5810

EFFECTIVE DATE: 01/27/94

APPROVED BY:

THOMAS L. CHARRON, M.D., M.P.H., SOLANO COUNTY HEALTH OFFICER, EMS MEDICAL DIRECTOR

REVIEWED BY:

ALLEN MORINI, D.O., ASSISTANT EMS MEDICAL DIRECTOR

SUBJECT: EMS AIRCRAFT – MEDICAL CONTROL

AUTHORITY: TITLE 22, CHAPTER 6, § 100206 – 100228;
CALIFORNIA HEALTH & SAFETY CODE § 1798.170

PURPOSE/POLICY:

To establish policy for medical control and destination decisions for EMS Aircraft from on-scene arrival until the patient is taken to a receiving hospital..

I. MEDICAL CONTROL:

A. Air ambulance aircraft

Air ambulance aircraft are staffed by personnel who have capabilities beyond that covered by Solano County protocols. Air ambulances authorized in Solano County are, in alphabetical order, (a) CALSTAR, and (b) REACH.

Medical Control for those procedures not covered by Solano County protocols is delegated to the Medical Director of the air ambulance service from the County Medical Director in accordance with the written agreement between the air ambulance provider and Solano County.

B. Rescue aircraft

On line medical control for ALS or BLS rescue aircraft will be the responsibility of the Solano County Base Hospital contacted by air or ground units. The ALS/BLS air rescue aircraft authorized in Solano County is Napa County CHP.

Treatment and destination decisions will be according to existing Solano County policy and protocols.

II. DESTINATION DECISIONS

- A. EMS Aircraft destination decisions will be initiated by the on-scene EMT-P (or aircraft crew) depending on who is on scene and in contact with the Solano County Base Hospital utilizing an appropriate recorded MED-COM radio channel.

For those cases transported to the closest appropriate facility with a helipad; those helipads are NorthBay Medical Center, VacaValley Hospital, or Sutter-Solano Medical Center. The Base Hospital will determine which is most appropriate in concert with ground and air crew.

- B. Patients requiring specialty care not available at the closest facility may be directed by the Base Hospital Physician to the closest specialty care center (i.e. Trauma Center). Base Hospital personnel shall notify the receiving facility immediately.

III. QUALITY ASSURANCE

The patient care forms, retrospective review mechanisms, unusual incident reports and other review systems will be the same as for all ground personnel as referenced in Solano County Policy #5820. All cases where air ambulances were used shall be reviewed for appropriateness of use of air ambulance resources, and care of patients. All EMS Aircraft shall submit a copy of their medical record(s) to the EMS Office as agreed to in the individual memorandum(s) of agreement.

ATTACHMENT 1

SPECIALIZED CRITICAL CARE CENTERS WITH HELIPAD CAPABILITY WITHIN 100 MILES OF SOLANO COUNTY

LEVEL 1 OR 2 TRAUMA CENTERS	City	Telephone Number
U. C. Davis Medical Center	Sacramento, CA	916-734-3797
John Muir Medical Center	Walnut Creek, CA	510-939-5800
San Jose Hospital	San Jose, CA	408-977-4444
Santa Clara Valley Medical Center	San Jose, CA	408-299-5311
Stanford University Medical Center	Palo Alto, CA	415-723-5631

PEDIATRIC TRAUMA CENTERS	City	Telephone Number
U. C. Davis Medical Center	Sacramento, CA	916-453-2994
Children's Hospital of Oakland	Oakland, CA	415-428-3240
P. I. C. U.		800-ICU-KIDS
Santa Clara Valley Medical Center	San Jose, CA	800-299-5301
Stanford University Medical Center	Palo Alto, CA	415-723-5494

BURN CENTER(S)	City	Telephone Number
Eden Hospital Burn Center	Castro Valley, CA	415-889-5066
Brookside Hospital Burn Center	San Pablo, CA	510-235-7000, ext 2784
U. C. Davis Medical Center Burn Center	Sacramento, CA	916-453-3636
Santa Clara Valley Med Ctr Burn Unit	San Jose, CA	408-299-5242

SPINAL CORD INJURY CENTER(S)	City	Telephone Number
No. CA. Regional Spinal Cord Injury Unit, Santa Clara Valley Med Ctr.	San Jose, CA	408-299-5143
U. C. Davis Medical Center	Sacramento, CA	916-453-3797

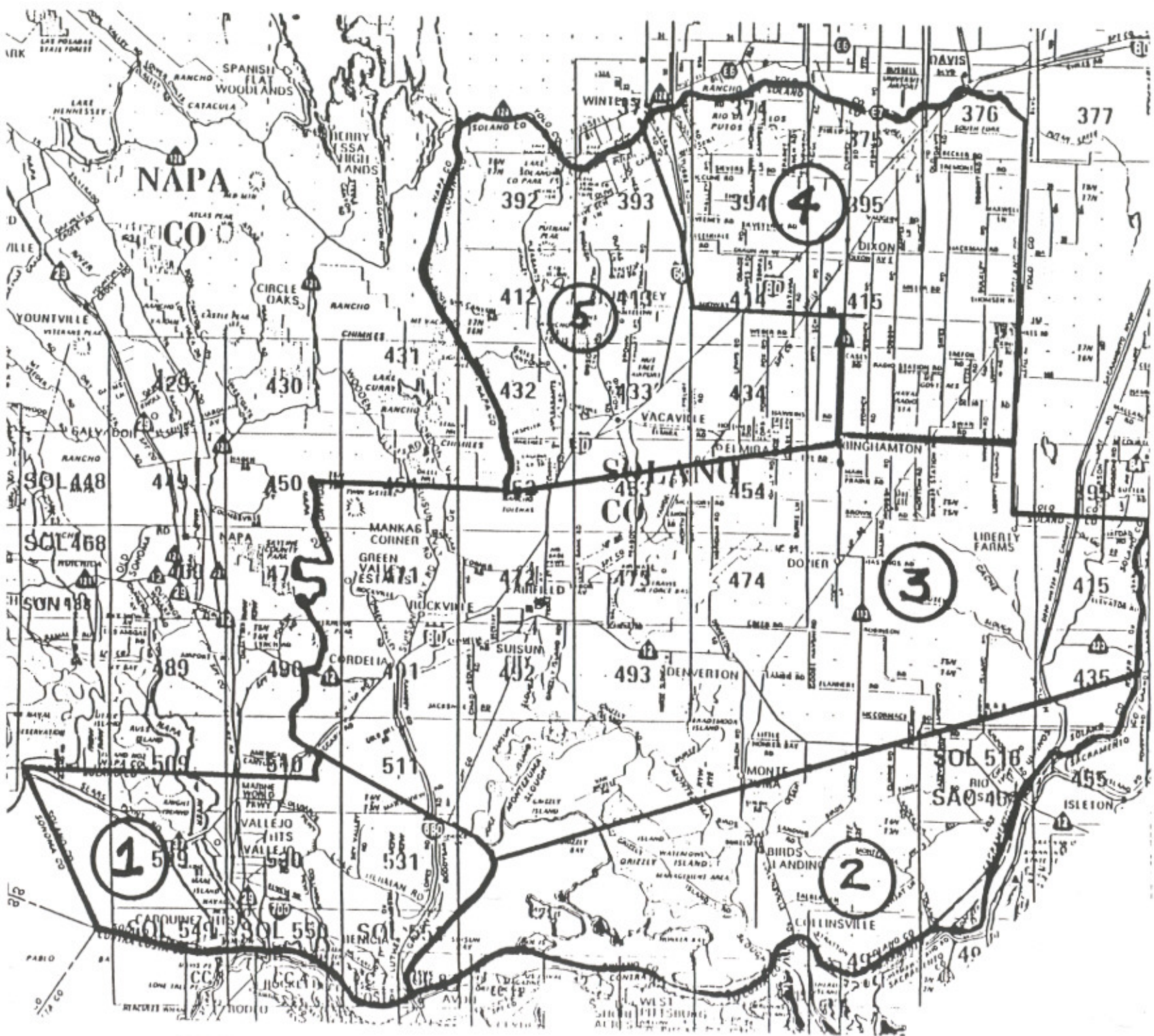
ATTACHMENT 2

Map attached indicating closest hospital with helipad capability for Solano EMS Patients needing transport by EMS Aircraft. Destination may be inappropriate if Base Hospital physician determines Base Hospital is unable to appropriately meet the patient's immediate medical needs.

ZONE	HOSPITAL
Zone 1	Sutter-Solano Medical Center, Vallejo
Zone 2	John Muir Medical Center, Walnut Creek, CA
Zone 3	North Bay Medical Center, Fairfield, CA
Zone 4	U. C. Davis Medical Center, Sacramento, CA
Zone 5	VacaValley Hospital, Vacaville, CA

SUBJECT: EMS Aircraft - Medical Control

Policy: 5810
Date: 05/01/92





Health Services Department

Public Health Division

Donald R. Rowe, Director

Thomas L. Charron, M.D., M.
Health Officer
Assistant Director

EMERGENCY MEDICAL SERVICES

POLICY MEMORANDUM #5820

EFFECTIVE DATE: 02/01/92

APPROVED BY:

THOMAS L. CHARRON, M.D., M.P.H., SOLANO COUNTY HEALTH OFFICER, EMS MEDICAL DIRECTOR

REVIEWED BY:

STEVEN J. RAMIREZ, M.D., F.A.C.E.P., ASSISTANT EMS MEDICAL DIRECTOR

SUBJECT: EMS AIRCRAFT AUTHORIZATION

AUTHORITY: CALIFORNIA CODE OF REGULATIONS, TITLE 22, CHAPTER 6, § 100206 – 100228;

PURPOSE/POLICY:

To establish authorization, training, orientation, staffing and safety policies for EMS aircraft.

I. AUTHORIZATION PROCEDURE

- A. The air ambulance/rescue service must submit:
 1. Application for initial authorization.
 2. A copy of its air taxi/commercial operators certificate. Public Safety agencies are exempt from certification requirements, but must document consistency with FAA requirements.
 3. Copy of its certificate of insurance policy.
 4. Proof of authorization and classification of the air ambulance/rescue provider by its local EMS agency of jurisdiction.

- B. Air ambulance services will assure that patient records contain data elements consistent with Article Six (100164) of Division 9, Title 22 of the California Administrative Code and submit documentation of patient care records of all EMS calls requesting the air ambulance services when the patient originates in Solano County.
- C. The air ambulance/rescue service will be required to have a written agreement with the local EMS agency to participate in the advanced life support program and to comply with all applicable state regulations, regional and local policy procedures, including a mechanism to assure compliance. The EMS Agency shall review its agreement with the authorized air ambulance service at least every two years. Such agreement may be changed, renewed, cancelled, or otherwise modified at that time. Authorization is dependent on maintaining a written agreement. The local EMS agency may deny, suspend, or revoke an air ambulance authorization for failure to comply with applicable policies, procedures and regulations. The local EMS Agency, at its discretion, shall have access to on-site inspection and appropriate record review of any authorized EMS aircraft provider as appropriate for unusual incident inquiries.
- D. Upon successful completion of A – C, the local EMS Agency will issue a certificate of authorization to the air ambulance/rescue service.
- E. The air ambulance/rescue service shall be designated by the local EMS agency as an approved air ambulance for the purpose of scene rescue, and ALS pre-hospital care. Except for specifically addressed EMS aircraft policy and procedures, all Solano EMS policies and procedures shall apply to its operation while operating in Solano County.

II. ORIENTATION OF EMS AIRCRAFT, PILOT AND CREW

The local EMS Agency shall assure that pilots and medical crews of authorized EMS aircraft are oriented to the local EMS systems before they operate within its jurisdiction. Each pilot and medical crew, prior to being utilized on an EMS mission within the local EMS agency, shall have completed an orientation approved by the EMS Agency.

- A. Pilot requirements – Orientation includes familiarization with:
 - 1. Landing sites and all pre-determined emergency landing sites and special operational information for each site within Solano County.
 - 2. A comprehensive communications inventory with frequency numbers, agency names and identifiers, PL Codes, etc., and special communications procedures of Solano County organizations and agencies that would interact with EMS Aircraft.
- B. Medical Crew – Orientation includes familiarization with:
 - 1. Pilot orientation requirements;
 - 2. Local medical control policies and procedures including but not limited to:

- a. "On line" medical direction;
- b. Record keeping requirements;
- c. Patient transfer between ground and air medical personnel.
- d. Patient destination.

III. AIR AMBULANCE MEDICAL FLIGHT CREW TRAINING AND STAFFING

- A. Air Ambulance means any aircraft specially constructed, modified or equipped and used for the primary purposes of responding to emergency calls and transporting critically ill or injured patients.
- B. Air Ambulance Flight Crew shall be a minimum of two (2) attendants certified or licensed in advanced life support. Prior to being utilized on any EMS mission, the crew members shall have training in aeromedical transportation as specified in the California Code of Regulations, Section 100320.

IV. AIR RESCUE MEDICAL FLIGHT CREW TRAINING AND STAFFING

- A. ALS Rescue Aircraft. ALS rescue aircraft shall have at least one (1) attendant who is certified at the advanced life support level (EMT-II, EMT-P). Prior to being utilized on any EMS mission, the crew member shall have training in aeromedical transportation as specified in the California Code of Regulations, Section 100320.
- B. BLS Rescue Aircraft: Shall have at least one (1) attendant who is certified as an EMT-I. Prior to being utilized on any EMS mission the crew member shall have training in aeromedical transportation as specified in the California Code of Regulations, Section 100320.

V. SAFETY POLICIES FOR AUTHORIZED EMS AIRCRAFT

Air ambulance/rescue service providers shall agree to comply with FAR Part 91 and 135 (or their equivalent) and AAMS Guidelines to maintain their authorization status, as well as all applicable State and Federal regulations. In addition, all safety policies will be adhered to as addressed in FAA Regulations.

- A. Each EMS authorized aircraft shall have the following equipment:
 - 1. A searchlight (manipulated by the pilot and having a minimum movement of 90 degrees vertical and 180 degrees horizontal) capable of illuminating a proposed landing site. It should be capable of operation by the pilot without requiring the pilot to remove his hands from the controls (NOT REQUIRED FOR DAY-VFR). It is recommended that searchlights have a minimum of 1 million candle power output for night operations.

2. Air to ground communications with those personnel first on the scene (i.e., POLICE, AMBULANCE, FIRE, etc.).
3. Additional restraining devices used to preclude patient interference with the flight crew or aircraft flight controls;
4. An intercommunications system for flight crew member and medical flight attendant communication aboard the aircraft.



Health Services Department

Public Health Division

Donald R. Rowe, Director

Thomas L. Charron, M.D., M.
Health Officer
Assistant Director

EMERGENCY MEDICAL SERVICES

POLICY MEMORANDUM #5830

EFFECTIVE DATE: 02/01/92

APPROVED BY:

THOMAS L. CHARRON, M.D., M.P.H., SOLANO COUNTY HEALTH OFFICER, EMS MEDICAL DIRECTOR

REVIEWED BY:

STEVEN G. KAHLINSKI, M.D., F.A.C.E.P., ASSISTANT EMS MEDICAL DIRECTOR

SUBJECT: TRANSFER OF EMS PATIENTS TO/FROM EMS AIRCRAFT

AUTHORITY: CALIFORNIA CODE OF REGULATIONS, TITLE 22, CHAPTER 6, § 100206 – 100228; CALIFORNIA HEALTH & SAFETY CODE 1798.170.

PURPOSE/POLICY:

To establish guidelines for the safe transfer of ground EMS patients to air EMS personnel.

I. GROUND/AIR TRANSFER

Ground EMS crews may transfer a patient to the flight crew of an EMS Rescue Aircraft when it is determined that rapid transport is a prime therapeutic intervention for the patient. It is the primary responsibility of the dispatch center to dispatch the most highly trained and staffed available EMS aircraft when requested.

II. MOST QUALIFIED PERSONNEL

If it is determined that EMS Rescue Aircraft personnel are less medically qualified than ground personnel, the appropriate base hospital will be contacted to determine whether or not the EMS rescue flight crew is capable of transporting the patient..

III. GUIDELINES WITHOUT BASE CONTACT

If base hospital contact is impossible, the following guidelines shall assist in the decision-making process whether or not the patient is transported by air.

- A. EMT-IIs on Air ALS rescue aircraft are capable of monitoring but not initiating field treatment provided by EMT-Ps or MICNs.
- B. EMT-Is on BLS rescue aircraft are capable of monitoring but not initiating field treatment provided by ALS ground crews.
- C. EMS rescue aircraft may allow ground EMS providers to be transported with the patient. The EMS provider going with the aircraft is considered a passenger assisting the patient and is not a member of the flight crew.
- D. Air/Ground Refusal – With Base Hospital contact and direction, an Air Ambulance crew may deem the patient(s) unacceptable for air transport, and may turn care back over to the ground crew.